GenomeAI — Genetic Analysis Report

Gene: BRCA1

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This report is generated from placeholder/mock data for development purposes and is not a clinical diagnosis.

Gene Validation

BRCA1 — BRCA1 DNA repair associated

Genetic Risk Assessment

Disease	Risk Level	Probability	Confidence
Hereditary Breast and Ovarian Cancer Syndrome	High	72%	90%
Fanconi Anemia, complementation group S	Low	8%	68%

Preventive Guidance

• [screening] Discuss enhanced screening frequency and imaging modality with a specialist.

• [lifestyle] General cancer-risk-reduction lifestyle factors (diet, exercise, no smoking) apply.

• [management] Genetic counseling is recommended to discuss risk-reducing options.

• [management] Discuss this result with a genetic counselor or physician before acting on it.

Explanation

Hereditary Breast and Ovarian Cancer Syndrome: BRCA1 was associated with Hereditary Breast and Ovarian Cancer Syndrome. This was categorized as high risk (72% estimated probability, 90% model confidence). The association was cross-checked against NCBI, ClinVar, OMIM and classified as pathogenic.

Fanconi Anemia, complementation group S: BRCA1 was associated with Fanconi Anemia, complementation group S. This was categorized as low risk (8% estimated probability, 68% model confidence). The association was cross-checked against NCBI, ClinVar, OMIM and classified as uncertain significance.

Supporting Research

BRCA1 mutation spectrum and cancer risk stratification — J. Clin. Oncol. (2025)

Population screening outcomes for hereditary cancer genes — Nature Medicine (2025)